

Faculdade de Ciências - Departamento de Computação

Disciplina: Métodos Numéricos Computacionais

Docente: Douglas Rodrigues

Lista de Exercícios - L3

Sistemas de Equações Lineares

Instruções:

- A lista é individual.
- A resolução deve ser feita de próprio punho (manuscrita).
- A lista deve ser entregue no formato PDF.
- O arquivo deve ser renomeado com o nome e sobrenome seguidos pela letra “L” e o número da lista.
- Deve-se utilizar letras minúsculas separados por traço baixo (underline).
- **Exemplo:** douglas_rodrigues_l3.pdf

Exercícios

1: Seja:

$$A = \begin{pmatrix} 16 & 4 & 8 & 4 \\ 4 & 10 & 8 & 4 \\ 8 & 8 & 12 & 10 \\ 4 & 4 & 10 & 12 \end{pmatrix}, \quad b = \begin{pmatrix} 32 \\ 26 \\ 38 \\ 30 \end{pmatrix}$$

- Calcule o determinante de A por cofatores.
- Verifique se A pode ser decomposta em LU. Se possível resolva o sistema por Decomposição LU.
- Calcule o determinante de A utilizando o teorema da Decomposição LU.

2: Seja:

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 0 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 9 \\ 1 \\ 7 \end{pmatrix}$$

- Verifique se a matriz A satisfaz as condições da Decomposição LU. Se sim, decomponha A em LU.
- Calcule o determinante de A utilizando o Teorema da Decomposição LU.

c) Resolva o sistema $Ax = b$.

3: Considere a matriz quadrada A com todas as submatrizes principais não singulares. Determine as fórmulas da Decomposição LU, onde L é matriz triangular inferior e U é matriz triangular superior com valor 1 na diagonal principal.

4: Resolva o sistema $Ax = b$ utilizando as fórmulas determinadas pelo exercício anterior, sendo:

$$A = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 0 & 2 \\ 0 & 3 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$$

5: Usando o Método de Gauss-Compacto resolva os sistemas:

a)

$$\begin{pmatrix} 2 & -1 & 3 \\ 4 & 1 & 2 \\ 1 & 0 & 10 \end{pmatrix} \begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{pmatrix} = \begin{pmatrix} -4 & 2 & 4 \\ -7 & 6 & 6 \\ -11 & 2 & 20 \end{pmatrix}$$

b)

$$\begin{aligned} 10x_1 + x_2 - x_3 &= 10 \\ x_1 + 10x_2 + x_3 &= 12 \\ 2x_1 - x_2 + 10x_3 &= 11 \end{aligned}$$

6: Seja:

$$A = \begin{pmatrix} 5 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$$

a) Verifique se A pode ser decomposta em LL^T . Se possível decomponha A .

b) Calcule o determinante de A pelo Teorema do Método de Cholesky.

c) Resolva o sistema $Ax = b$.

7: Resolva o sistema linear a seguir pelo Método de Gauss-Jordan:

a)

$$\begin{aligned} x_1 + 2x_2 + 4x_3 + x_4 &= 1 \\ 2x_1 + x_2 - x_3 + 3x_4 &= -6 \\ -x_1 + 3x_2 + x_3 - 5x_4 &= 7 \\ 3x_1 + 4x_2 + 2x_3 - 6x_4 &= 13 \end{aligned}$$

b)

$$\begin{aligned} 2x_1 + 4x_2 + 6x_3 + x_4 &= -8 \\ x_1 - 3x_2 - x_3 + 2x_4 &= 4 \\ 2x_1 + x_2 + 3x_3 - x_4 &= -2 \\ 4x_1 - 2x_2 + x_3 - 4x_4 &= 6 \end{aligned}$$

8: Usando o Método de Cholesky, calcule A^{-1} :

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 10 & 2 \\ -1 & 2 & 4 \end{pmatrix}$$

9: Calcule A^{-1} utilizando o Método de Gauss-Compacto:

a)

$$A = \begin{pmatrix} 4 & 2 & -4 \\ 2 & 10 & 4 \\ -4 & 4 & 9 \end{pmatrix}$$

b)

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & -3 & -1 \\ 2 & 1 & 1 \end{pmatrix}$$

10: Usando o Método da Decomposição LU, calcule A^{-1} :

a)

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & -2 & 3 & 4 \\ 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

b)

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 10 & 2 \\ -1 & 2 & 4 \end{pmatrix}$$

11: Verifique a convergência para o Método de Jacobi e, se possível, resolva os sistemas lineares:

a)

$$5x_1 + 2x_2 + x_3 = 7$$

$$-x_1 + 4x_2 + 2x_3 = 3$$

$$2x_1 - 3x_2 + 10x_3 = -1$$

$$x^{(0)} = \begin{pmatrix} -1.4 \\ 3.9 \\ 0.2 \end{pmatrix}, \quad \varepsilon = 10^{-1}$$

b)

$$10x_1 + x_2 + x_3 = 10$$

$$2x_1 + 10x_2 + 8x_3 = 20$$

$$7x_1 + x_2 + 10x_3 = 20$$

$$x^{(0)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \varepsilon = 10^{-2}$$

12: Verifique a convergência para o Método de Gauss-Seidel e, se possível, resolva os sistemas lineares:

a)

$$\begin{aligned}4x_1 + x_2 + x_3 + x_4 &= 7 \\2x_1 - 8x_2 + x_3 - x_4 &= -6 \\x_1 + 2x_2 - 5x_3 + x_4 &= -1 \\x_1 + x_2 + x_3 - 4x_4 &= -1\end{aligned}$$

$$x^{(0)} = \begin{pmatrix} 0 \\ 0.2 \\ -0.4 \\ 0 \end{pmatrix}, \quad \varepsilon = 10^{-2}$$

b)

$$\begin{aligned}4x_1 + x_2 + x_3 &= 6 \\x_1 + 6x_2 + x_3 &= 8 \\2x_1 + x_2 + 8x_3 &= 11\end{aligned}$$

$$x^{(0)} = \begin{pmatrix} 0 \\ 0.1 \\ 0 \end{pmatrix}, \quad \varepsilon = 10^{-2}$$

Lista de Exercícios 3 - Sistema de Equações Lineares

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1- $A = \begin{pmatrix} 16 & 4 & 8 & 4 \\ 4 & 10 & 8 & 4 \\ 8 & 8 & 12 & 10 \\ 4 & 4 & 10 & 12 \end{pmatrix}, b = \begin{pmatrix} 32 \\ 26 \\ 38 \\ 30 \end{pmatrix}$

$$\det A = 16 \cdot (-1)^{11} \cdot \det \begin{vmatrix} 10 & 8 & 4 \\ 8 & 12 & 10 \\ 4 & 10 & 12 \end{vmatrix} + 4 \cdot (-1)^{12} \cdot \det \begin{vmatrix} 4 & 8 & 4 \\ 8 & 12 & 10 \\ 4 & 10 & 12 \end{vmatrix} + 8 \cdot (-1)^{13} \cdot \det \begin{vmatrix} 4 & 10 & 4 \\ 8 & 8 & 10 \\ 4 & 4 & 12 \end{vmatrix} + 4 \cdot (-1)^{14} \cdot \det \begin{vmatrix} 4 & 10 & 8 \\ 8 & 8 & 12 \\ 4 & 4 & 10 \end{vmatrix}$$

$$\det a = 10 \cdot (-1)^{11} \cdot \det \begin{vmatrix} 12 & 10 \\ 10 & 12 \end{vmatrix} + 8 \cdot (-1)^{12} \cdot \det \begin{vmatrix} 8 & 10 \\ 4 & 12 \end{vmatrix} + 4 \cdot (-1)^{13} \cdot \det \begin{vmatrix} 8 & 12 \\ 4 & 10 \end{vmatrix}$$

$$\det a_1 = 12 \cdot (-1)^{11} \cdot 12 + 10 \cdot (-1)^{12} \cdot 10 = 44$$

$$\det a_2 = 8 \cdot (-1)^{11} \cdot 12 + 10 \cdot (-1)^{12} \cdot 4 = 56$$

$$\det a_3 = 8 \cdot (-1)^{11} \cdot 10 + 12 \cdot (-1)^{12} \cdot 4 = 32$$

$$\det a = 10 \cdot (-1)^2 \cdot 44 + 8 \cdot (-1)^3 \cdot 56 + 4 \cdot (-1)^4 \cdot 32 = 120$$

$$\det b = 4 \cdot (-1)^2 \cdot \det \begin{vmatrix} 12 & 10 \\ 10 & 12 \end{vmatrix} + 8 \cdot (-1)^3 \cdot \det \begin{vmatrix} 8 & 10 \\ 4 & 12 \end{vmatrix} + 4 \cdot (-1)^4 \cdot \det \begin{vmatrix} 8 & 12 \\ 4 & 10 \end{vmatrix}$$

$$\det b = 4 \cdot (-1)^2 \cdot 44 + 8 \cdot (-1)^3 \cdot 56 + 4 \cdot (-1)^4 \cdot 32 = -144$$

$$\det c = 4 \cdot (-1)^2 \cdot \det \begin{vmatrix} 8 & 10 \\ 4 & 12 \end{vmatrix} + 10 \cdot (-1)^3 \cdot \det \begin{vmatrix} 8 & 10 \\ 4 & 12 \end{vmatrix} + 4 \cdot (-1)^4 \cdot \det \begin{vmatrix} 8 & 8 \\ 4 & 4 \end{vmatrix}$$

$$\det c_1 = 8(-1)^2 \cdot 12 + 10(-1)^3 \cdot 4 = 56$$

$$\det c_2 = c_1 = 56$$

$$\det c_3 = 8(-1)^2 \cdot 4 + 8(-1)^3 \cdot 4 = 0$$

$$\det c = 4(-1)^2 \cdot 56 + 10(-1)^3 \cdot 56 + 4(-1)^4 \cdot 0 = -336$$

$$\det d = 4(-1)^2 \overbrace{\begin{vmatrix} 8 & 12 \\ 4 & 10 \end{vmatrix}}^{d_1} + 10(-1)^3 \overbrace{\begin{vmatrix} 8 & 12 \\ 4 & 10 \end{vmatrix}}^{d_2} + 8(-1)^4 \overbrace{\begin{vmatrix} 8 & 8 \\ 4 & 4 \end{vmatrix}}^{c_3}$$

$$\det d_1 = 8(-1)^2 \cdot 10 + 12(-1)^3 \cdot 4 = 32$$

$$\det d_2 = d_1 = 32$$

$$\det d = 4(-1)^2 \cdot 32 + 10(-1)^3 \cdot 32 + 8(-1)^4 \cdot 0 = -192$$

$$\det A = 16(-1)^{11} \cdot 120 + 4(-1)^{1+2}(-144) + 8(-1)^{1+3}(-336) + 4(-1)^{1+4}(-192)$$

$$\det A = 576 //$$

b) $\det(A_k) \neq 0$, $k=1, 2, 3, 4$

$$|A_1| = 16 \neq 0$$

$$|A_2| = 16(-1)^2 \cdot 10 + 4(-1)^3 \cdot 4 = 144 \neq 0$$

$$|A_3| = 16(-1)^2 \overbrace{\begin{vmatrix} 10 & 8 \\ 8 & 12 \end{vmatrix}}^{A_{31}} + 4(-1)^3 \overbrace{\begin{vmatrix} 4 & 8 \\ 8 & 12 \end{vmatrix}}^{A_{32}} + 8(-1)^4 \overbrace{\begin{vmatrix} 4 & 10 \\ 8 & 8 \end{vmatrix}}^{A_{33}}$$

$$|A_{31}| = 10(-1)^2 \cdot 12 + 8(-1)^3 \cdot 8 = 56$$

$$|A_{32}| = 4(-1)^2 \cdot 12 + 8(-1)^3 \cdot 8 = -16$$

$$|A_{33}| = 4(-1)^2 \cdot 8 + 10(-1)^3 \cdot 8 = -48$$

$$|A_3| = 16(-1)^2 \cdot 56 + 4(-1)^3 \cdot (-16) + 8(-1)^4 \cdot (-48) = -96 \neq 0$$

$$|A_4| = 576 \neq 0$$

$\therefore$ converge

$$L = \begin{pmatrix} 1 & & & \\ L_{21} & 1 & & \\ L_{31} & L_{32} & 1 & \\ L_{41} & L_{42} & L_{43} & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} \mu_{11} & \mu_{12} & \mu_{13} & \mu_{14} \\ & \mu_{22} & \mu_{23} & \mu_{24} \\ & & \mu_{33} & \mu_{34} \\ & & & \mu_{44} \end{pmatrix}$$

$$\begin{pmatrix} \mu_{11} & \mu_{12} & \mu_{13} & \mu_{14} \\ L_{21}\mu_{11} & L_{21}\mu_{12} + \mu_{22} & L_{21}\mu_{13} + \mu_{23} & L_{21}\mu_{14} + \mu_{24} \\ L_{31}\mu_{11} & L_{31}\mu_{12} + L_{32}\mu_{22} & L_{31}\mu_{13} + L_{32}\mu_{23} + \mu_{33} & L_{31}\mu_{14} + L_{32}\mu_{24} + \mu_{34} \\ L_{41}\mu_{11} & L_{41}\mu_{12} + L_{42}\mu_{22} & L_{41}\mu_{13} + L_{42}\mu_{23} + L_{43}\mu_{33} & L_{41}\mu_{14} + L_{42}\mu_{24} + L_{43}\mu_{34} + \mu_{44} \end{pmatrix} = \begin{pmatrix} 16 & 4 & 8 & 4 \\ 4 & 10 & 8 & 4 \\ 8 & 8 & 12 & 10 \\ 4 & 4 & 10 & 12 \end{pmatrix}$$

$$\mu_{11} = 16, \mu_{12} = 4, \mu_{13} = 8, \mu_{14} = 4$$

$$L_{21}16 = 4 \rightarrow L_{21} = \frac{1}{4}, L_{31}16 = 8 \rightarrow L_{31} = \frac{1}{2}, L_{41}16 = 4 \rightarrow L_{41} = \frac{1}{4}$$

$$\frac{1}{4} \cdot 4 + \mu_{22} = 10 \rightarrow \mu_{22} = 9, \frac{1}{4} \cdot 8 + \mu_{23} = 8 \rightarrow \mu_{23} = 6, \frac{1}{4} \cdot 4 + \mu_{24} = 4 \rightarrow \mu_{24} = 3$$

$$\frac{1}{2} \cdot 4 + L_{32} \cdot 9 = 8 \rightarrow L_{32} = \frac{2}{3}, \frac{1}{2} \cdot 8 + \frac{2}{3} \cdot 6 + \mu_{33} = 12 \rightarrow \mu_{33} = 4, \frac{1}{2} \cdot 4 + \frac{2}{3} \cdot 3 + \mu_{34} = 10 \rightarrow \mu_{34} = 6$$

$$\frac{1}{4} \cdot 4 + L_{42} \cdot 9 = 4 \rightarrow L_{42} = \frac{1}{3}, \frac{1}{4} \cdot 8 + \frac{1}{3} \cdot 6 + L_{43} \cdot 4 = 10 \rightarrow L_{43} = \frac{3}{2}, \frac{1}{4} \cdot 4 + \frac{1}{3} \cdot 3 + \frac{3}{2} \cdot 6 + \mu_{44} = 12 \rightarrow \mu_{44} = 1$$

$$L = \begin{pmatrix} 1 & & & \\ \frac{1}{4} & 1 & & \\ \frac{1}{2} & \frac{2}{3} & 1 & \\ \frac{1}{4} & \frac{1}{3} & \frac{3}{2} & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 16 & 4 & 8 & 4 \\ & 9 & 6 & 3 \\ & & 4 & 6 \\ & & & 1 \end{pmatrix}$$

$$Ly = b$$

$$\begin{pmatrix} 1 & & & \\ \frac{1}{4} & 1 & & \\ \frac{1}{2} & \frac{2}{3} & 1 & \\ \frac{1}{4} & \frac{1}{3} & \frac{3}{2} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 32 \\ 26 \\ 38 \\ 30 \end{pmatrix} \rightarrow \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 32 \\ 18 \\ 10 \\ \frac{28}{3} \end{pmatrix}$$

$$Ux = y$$

$$\begin{pmatrix} 16 & 4 & 8 & 4 \\ & 9 & 6 & 3 \\ & & 4 & 6 \\ & & & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 32 \\ 18 \\ 10 \\ \frac{28}{3} \end{pmatrix} \rightarrow x = \begin{pmatrix} \frac{34}{9} \\ \frac{59}{9} \\ -\frac{23}{2} \\ \frac{28}{3} \end{pmatrix}$$

$$c) \det A = \det L \cdot \det U = 576$$

$$\det L = 1$$

$$\det U = 16 \cdot 9 \cdot 4 \cdot 1 = 576$$

$$2- \quad A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 0 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 9 \\ 1 \\ 7 \end{pmatrix}$$

$$a) \quad |A_1| = 2 \neq 0$$

$$|A_2| = 2(-1)^2 \cdot (-1) + 1(-1)^3 \cdot 0 = -2 \neq 0$$

$$|A_3| = 1(-1)^2 \overbrace{\begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix}}^{A_{31}} + 3(-1)^4 \overbrace{\begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix}}^{A_{32}} = 1 \cdot 4 + 3(-2) = -2 \neq 0$$

$$A_{31} = 1(-1)^2 \cdot 1 + 3(-1)^3(-1) = 4$$

$$A_{32} = 2(-1)^2(-1) = -2$$

Portante, satisfaz o teorema

$$L = \begin{pmatrix} 1 & & \\ L_{21} & 1 & \\ L_{31} & L_{32} & 1 \end{pmatrix}, \quad U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{pmatrix}$$

$$\begin{pmatrix} u_{11} & u_{12} & u_{13} \\ L_{21}u_{11} & L_{21}u_{12} + u_{22} & L_{21}u_{13} + u_{23} \\ L_{31}u_{11} & L_{31}u_{12} + L_{32}u_{22} & L_{31}u_{13} + L_{32}u_{23} + u_{33} \end{pmatrix} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

$$u_{11} = 2, \quad u_{12} = 1, \quad u_{13} = 3,$$

$$L_{21}u_{11} = 0 \rightarrow L_{21} = 0, \quad L_{21}u_{12} + u_{22} = -1 \rightarrow u_{22} = -1, \quad L_{21}u_{13} + u_{23} = 1 \rightarrow u_{23} = 1,$$

$$L_{31}u_{11} = 1 \rightarrow L_{31} = \frac{1}{2}, \quad L_{31}u_{12} + L_{32}u_{22} = 0 \rightarrow \frac{1}{2} \cdot 1 + L_{32}(-1) = 0 \rightarrow L_{32} = \frac{1}{2},$$

$$\frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 1 + u_{33} = 3 \rightarrow u_{33} = 1,$$

$$b) \det A = \det L \cdot \det U$$

$$U = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

$$\det U = 1(-1)^6 \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} = 1(-1)^6 \cdot (-1)(-1)^4 \cdot 2 = -2$$

$$\det L = 1(-1)^2 \begin{vmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{vmatrix} = 1 \cdot 1(-1)^2 \cdot 1 = 1$$

$$\det A = 1 \cdot (-2) = -2 //$$

c)

$$Ly = b$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 9 \\ 1 \\ 7 \end{pmatrix}$$

$$y_1 = 9 \quad \frac{1}{2}y_1 + \frac{1}{2}y_2 + y_3 = 7$$

$$y_2 = 1 \quad y_3 = 2$$

$$y = (9 \ 1 \ 2)$$

$$Ux = y$$

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 9 \\ 1 \\ 2 \end{pmatrix}$$

$$2x_1 + x_2 + 3x_3 = 9 \rightarrow 2x_1 + 1 + 6 = 9 \rightarrow x_1 = 1$$

$$-x_2 + x_3 = 1 \rightarrow x_2 = 1$$

$$x_3 = 2$$

$$x = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} //$$

3-

$$L = \begin{pmatrix} l_{11} & 0 & \dots & 0 \\ l_{21} & l_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{m1} & l_{m2} & \dots & l_{mm} \end{pmatrix}, \quad U = \begin{pmatrix} 1 & u_{12} & \dots & u_{1m} \\ 0 & 1 & \dots & u_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

$$l_{ij} = a_{ij} - \sum_{k=1}^{j-1} l_{ik} u_{kj}, \quad i \geq j //$$

$$\mu_{ij} = \frac{1}{l_{ii}} \left(a_{ij} - \sum_{k=1}^{i-1} l_{ik} \mu_{kj} \right), \quad i < j //$$

$$4- \quad A = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 0 & 2 \\ 0 & 3 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 3 & -1 \\ 0 & -3/2 & 5/2 \\ 0 & 0 & 4 \end{pmatrix}$$

$$Ly = b$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$$

$$y_1 = 4$$

$$1/2 y_1 + y_2 = 3 \rightarrow y_2 = 1$$

$$-2y_2 + y_3 = 2 \rightarrow y_3 = 4$$

$$y = (4 \ 1 \ 4)$$

$$Ux = b$$

$$\begin{pmatrix} 2 & 3 & -1 \\ 0 & -3/2 & 5/2 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 4 \end{pmatrix}$$

$$2x_1 + 3x_2 - x_3 = 4 \rightarrow x_1 = 1$$

$$-3/2 x_2 + 5/2 x_3 = 1 \rightarrow x_2 = 1$$

$$4x_3 = 4 \rightarrow x_3 = 1$$

$$x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} //$$

$$5- a) \quad \begin{pmatrix} 2 & -1 & 3 \\ 4 & 1 & 2 \\ 1 & 0 & 10 \end{pmatrix} \begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{pmatrix} = \begin{pmatrix} -4 & 2 & 4 \\ -7 & 6 & 6 \\ -11 & 2 & 20 \end{pmatrix}$$

$$\begin{pmatrix} \mu_{11} & \mu_{12} & \mu_{13} & \vdots & \mu_{14} & \mu_{15} & \mu_{16} \\ l_{21} & \mu_{22} & \mu_{23} & \vdots & \mu_{24} & \mu_{25} & \mu_{26} \\ l_{31} & l_{32} & \mu_{33} & \vdots & \mu_{34} & \mu_{35} & \mu_{36} \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 3 & \vdots & -4 & 2 & 4 \\ 2 & 3 & -4 & \vdots & 1 & 2 & -2 \\ 1/2 & 1/6 & 55/6 & \vdots & -55/6 & 2/3 & 55/3 \end{pmatrix}$$

Para x:

$$\begin{pmatrix} 2 & -1 & 3 \\ 0 & 3 & -4 \\ 0 & 0 & 55/6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -4 \\ 1 \\ -55/6 \end{pmatrix} \rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix} //$$

Para y:

$$\begin{pmatrix} 2 & -1 & 3 \\ 0 & 3 & -4 \\ 0 & 0 & 55/6 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2/3 \end{pmatrix} \rightarrow \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 4/11 \\ 42/55 \\ 4/55 \end{pmatrix} //$$

Para z:

$$\begin{pmatrix} 2 & -1 & 3 \\ 0 & 3 & -4 \\ 0 & 0 & 55/6 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ 55/3 \end{pmatrix} \rightarrow \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} //$$

b) $10x_1 + x_2 - x_3 = 10$
 $x_1 + 10x_2 + x_3 = 12$
 $2x_1 - x_2 + 10x_3 = 11$

$$\rightarrow \begin{pmatrix} 10 & 1 & -1 \\ 1 & 10 & 1 \\ 2 & -1 & 10 \end{pmatrix} \text{ e } \begin{pmatrix} 10 \\ 12 \\ 11 \end{pmatrix} \rightarrow \left(\begin{array}{ccc|c} 10 & 1 & -1 & 10 \\ 1 & 10 & 1 & 12 \\ 2 & -1 & 10 & 11 \end{array} \right)$$

$$\left(\begin{array}{ccc|c} 10 & 1 & -1 & 10 \\ 0 & 9,9 & 1,1 & 11 \\ 0 & -1,2 & 10,2 & 9 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 10 & 1 & -1 & 10 \\ 0 & 99/10 & 11/10 & 11 \\ 0 & 0 & 31/3 & 31/3 \end{array} \right)$$

$$\begin{pmatrix} 10 & 1 & -1 \\ 0 & 99/10 & 11/10 \\ 0 & 0 & 31/3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 10 \\ 11 \\ 31/3 \end{pmatrix} \rightarrow \begin{aligned} 10x_1 + x_2 - x_3 &= 10 \\ 99/10 x_2 + 11/10 x_3 &= 11 \\ 31/3 x_3 &= 31/3 \end{aligned} \quad x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} //$$

$$6- \quad A = \begin{pmatrix} 5 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$$

$$a) |A_1| = 5 > 0$$

$$|A_2| = 5 \cdot (-1)^2 \cdot 1 + 1 \cdot (-1)^3 \cdot 1 = 4 > 0$$

$$|A_3| = 5 \cdot (-1)^2 \cdot \underbrace{\begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix}}_{A_{31}} + 1 \cdot (-1)^3 \cdot \underbrace{\begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}}_{A_{32}} + 2 \cdot (-1)^4 \cdot \underbrace{\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}}_{A_{33}} = 10 - 1 - 2 = 7 > 0$$

$$A_{31} = 1 \cdot (-1)^2 \cdot 3 + 1 \cdot (-1)^3 \cdot 1 = 2$$

$$A_{32} = 1 \cdot (-1)^2 \cdot 3 + 1 \cdot (-1)^3 \cdot 2 = 1$$

$$A_{33} = 1 \cdot (-1)^2 \cdot 1 + 1 \cdot (-1)^3 \cdot 2 = -1$$

$$L = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix}; \quad L^T = \begin{pmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{pmatrix}$$

$$L \cdot L^T = \begin{pmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{pmatrix} = \begin{pmatrix} 5 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

$$l_{11} = \sqrt{5}, \quad l_{21} = \frac{1}{\sqrt{5}}, \quad l_{31} = \frac{3}{\sqrt{5}}, \quad l_{32} = \frac{3\sqrt{5}}{10}, \quad l_{33} = \frac{\sqrt{7}}{2}$$

$$L = \begin{pmatrix} \sqrt{5} & 0 & 0 \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} & 0 \\ \frac{2\sqrt{5}}{5} & \frac{3\sqrt{5}}{10} & \frac{\sqrt{7}}{2} \end{pmatrix}; \quad L^T = \begin{pmatrix} \sqrt{5} & \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \\ 0 & \frac{2\sqrt{5}}{5} & \frac{3\sqrt{5}}{10} \\ 0 & 0 & \frac{\sqrt{7}}{2} \end{pmatrix} //$$

$$b) \det A = (\det L)^2 \rightarrow \det A = (\sqrt{7})^2 \rightarrow \det A = 7 //$$

$$\det L = \sqrt{5} \cdot (-1)^2 \cdot \begin{vmatrix} \frac{2\sqrt{5}}{5} & 0 \\ \frac{3\sqrt{5}}{10} & \frac{\sqrt{7}}{2} \end{vmatrix} = \sqrt{5} \cdot \frac{2}{\sqrt{5}} \cdot (-1)^2 \cdot \frac{\sqrt{7}}{2} = \sqrt{7}$$

c) $Ly = b$

$$\begin{pmatrix} \sqrt{5} & 0 & 0 \\ \sqrt{5}/5 & 2\sqrt{5}/5 & 0 \\ 2\sqrt{5}/5 & 3\sqrt{5}/10 & \sqrt{7}/2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$$

$$\sqrt{5} y_1 = -1 \rightarrow y_1 = -\sqrt{5}/5$$

$$\frac{\sqrt{5}}{5} y_1 + \frac{2\sqrt{5}}{5} y_2 = 1 \rightarrow y_2 = \frac{3\sqrt{5}}{5}$$

$$\frac{-2}{5} + \frac{9}{10} + \frac{\sqrt{7}}{2} y_3 = 4 \rightarrow y_3 = \sqrt{7}$$

$$y = \begin{pmatrix} -\frac{\sqrt{5}}{5} & \frac{3\sqrt{5}}{5} & \sqrt{7} \end{pmatrix}$$

$$L^T x = y$$

$$\begin{pmatrix} \sqrt{5} & \sqrt{5}/5 & 2\sqrt{5}/5 \\ 0 & 2\sqrt{5}/5 & 3\sqrt{5}/10 \\ 0 & 0 & \sqrt{7}/2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -\sqrt{5}/5 \\ 3\sqrt{5}/5 \\ \sqrt{7} \end{pmatrix} \rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

7- a) $x_1 + 2x_2 + 4x_3 + x_4 = 1$

$$2x_1 + x_2 - x_3 + 3x_4 = -6$$

$$-x_1 + 3x_2 + x_3 - 5x_4 = 7$$

$$3x_1 + 4x_2 + 2x_3 - 6x_4 = 13$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 4 & 1 & 1 \\ 2 & 1 & -1 & 3 & -6 \\ -1 & 3 & 1 & -5 & 7 \\ 3 & 4 & 2 & -6 & 13 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 4 & 1 & 1 \\ 0 & -3 & -9 & 1 & -8 \\ 0 & 5 & 5 & -4 & 8 \\ 0 & -2 & -10 & -9 & 10 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 4 & 1 & 1 \\ 2 & 1 & 3 & -1/3 & 8/3 \\ -1 & 5 & 5 & -4 & 8 \\ 3 & -2 & -10 & -9 & 10 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 4 & 1 & 1 \\ 0 & 1 & 3 & -1/3 & 8/3 \\ 0 & 0 & -10 & -7/3 & -16/3 \\ 0 & 0 & -4 & -29/3 & 46/3 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 4 & 1 & 1 \\ 0 & 1 & 3 & -\frac{1}{3} & \frac{8}{3} \\ 0 & 0 & 1 & \frac{7}{30} & \frac{8}{15} \\ 0 & 0 & -4 & -\frac{29}{3} & \frac{46}{3} \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 4 & 1 & 1 \\ 0 & 1 & 3 & -\frac{1}{3} & \frac{8}{3} \\ 0 & 0 & 1 & \frac{7}{30} & \frac{8}{15} \\ 0 & 0 & 0 & -\frac{131}{15} & \frac{262}{15} \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 4 & 1 & 1 \\ 0 & 1 & 3 & -\frac{1}{3} & \frac{8}{3} \\ 0 & 0 & 1 & \frac{7}{30} & \frac{8}{15} \\ 0 & 0 & 0 & 1 & -2 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 4 & 0 & 3 \\ 0 & 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -2 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -2 \end{array} \right) \rightarrow x = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -2 \end{pmatrix}$$

$$b) \left(\begin{array}{cccc|c} 2 & 4 & 6 & 1 & -8 \\ 1 & -3 & -1 & 2 & 4 \\ 2 & 1 & 3 & -1 & -2 \\ 4 & -2 & 1 & -4 & 6 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & \frac{1}{2} & -4 \\ 1 & -3 & -1 & 2 & 4 \\ 2 & 1 & 3 & -1 & -2 \\ 4 & -2 & 1 & -4 & 6 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & \frac{1}{2} & -4 \\ 0 & -5 & -4 & \frac{3}{2} & 8 \\ 0 & -3 & -3 & -2 & 6 \\ 0 & -10 & -11 & -6 & 22 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & \frac{1}{2} & -4 \\ 0 & 1 & \frac{4}{5} & -\frac{3}{10} & -\frac{2}{5} \\ 0 & -3 & -3 & -2 & 6 \\ 0 & -10 & -11 & -6 & 22 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & \frac{1}{2} & -4 \\ 0 & 1 & \frac{4}{5} & -\frac{3}{10} & -\frac{2}{5} \\ 0 & 0 & -\frac{3}{5} & -\frac{29}{10} & \frac{6}{5} \\ 0 & 0 & -3 & -9 & 6 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & \frac{1}{2} & -4 \\ 0 & 1 & \frac{4}{5} & -\frac{3}{10} & -\frac{2}{5} \\ 0 & 0 & 1 & \frac{29}{6} & -2 \\ 0 & 0 & -3 & -9 & 6 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & \frac{1}{2} & -4 \\ 0 & 1 & \frac{4}{5} & -\frac{3}{10} & -\frac{8}{5} \\ 0 & 0 & 1 & \frac{29}{6} & -2 \\ 0 & 0 & 0 & \frac{11}{2} & 0 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & \frac{1}{2} & -4 \\ 0 & 1 & \frac{4}{5} & -\frac{3}{10} & -\frac{8}{5} \\ 0 & 0 & 1 & \frac{29}{6} & -2 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 3 & 0 & -4 \\ 0 & 1 & \frac{4}{5} & 0 & -\frac{8}{5} \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right) \rightarrow X = \begin{pmatrix} 2 \\ 0 \\ -2 \\ 1 \end{pmatrix}$$

8-

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 10 & 2 \\ -1 & 2 & 4 \end{pmatrix}$$

$$L = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix}; \quad L^T = \begin{pmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{pmatrix}$$

$$L \cdot L^T = \begin{pmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{pmatrix} = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 10 & 2 \\ -1 & 2 & 4 \end{pmatrix}$$

$$l_{11} = \sqrt{2} \quad l_{31} = -\frac{\sqrt{2}}{2} \quad l_{22} = \frac{\sqrt{19}}{2}$$

$$l_{21} = \frac{\sqrt{2}}{2} \quad l_{32} = \frac{5\sqrt{38}}{38} \quad l_{33} = \frac{\sqrt{1026}}{\sqrt{361}}$$

$$L_y = I$$

$$\begin{pmatrix} \sqrt{2} & 0 & 0 \\ \sqrt{2}/2 & \sqrt{19}/2 & 0 \\ -\sqrt{2}/2 & 5\sqrt{38}/38 & \sqrt{1026}/361 \end{pmatrix} \begin{pmatrix} y_{11} & y_{12} & y_{13} \\ y_{21} & y_{22} & y_{23} \\ y_{31} & y_{32} & y_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$y = \begin{pmatrix} \sqrt{2}/2 & -\sqrt{38}/38 & 2\sqrt{114}/57 \\ 0 & \sqrt{38}/19 & -5\sqrt{114}/342 \\ 0 & 0 & \sqrt{114}/18 \end{pmatrix}$$

$$L^T x = y$$

$$\begin{pmatrix} \sqrt{2} & \sqrt{2}/2 & -\sqrt{2}/2 \\ 0 & \sqrt{19}/2 & 5\sqrt{38}/38 \\ 0 & 0 & 3\sqrt{114}/19 \end{pmatrix} \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix} = \begin{pmatrix} \sqrt{2}/2 & -\sqrt{38}/38 & 2\sqrt{114}/57 \\ 0 & \sqrt{38}/19 & -5\sqrt{114}/342 \\ 0 & 0 & \sqrt{114}/18 \end{pmatrix}$$

$$x = \begin{pmatrix} 2/3 & -3/9 & 2/9 \\ -1/9 & 7/54 & -5/54 \\ 2/9 & -5/54 & 19/54 \end{pmatrix}$$

9- a)

$$A = \begin{pmatrix} 4 & 2 & -4 \\ 2 & 10 & 4 \\ -4 & 4 & 9 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 4 & 2 & -4 & 1 & 0 & 0 \\ 2 & 10 & 4 & 0 & 1 & 0 \\ -4 & 4 & 9 & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 4 & 2 & -4 & 1 & 0 & 0 \\ 0 & 9 & 6 & -1/2 & 1 & 0 \\ 0 & 6 & 5 & 1 & 0 & 1 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|ccc} 4 & 2 & -4 & 1 & 0 & 0 \\ 0 & 9 & 6 & -1/2 & 1 & 0 \\ 0 & 0 & 1 & 4/3 & -2/3 & 1 \end{array} \right)$$

1ª coluna:

$$\begin{pmatrix} 4 & 2 & -4 \\ 0 & 9 & 6 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{12} \\ a_{13} \end{pmatrix} = \begin{pmatrix} 1 \\ -1/2 \\ 4/3 \end{pmatrix} \rightarrow \begin{aligned} a_{11} &= 37/18 \\ a_{12} &= -17/18 \\ a_{13} &= 4/3 \end{aligned}$$

2ª coluna:

$$\begin{pmatrix} 4 & 2 & -4 \\ 0 & 9 & 6 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{21} \\ a_{22} \\ a_{23} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -2/3 \end{pmatrix} \rightarrow \begin{aligned} 4a_{21} + 2a_{22} - 4a_{23} &= 0 \rightarrow a_{21} = -17/18 \\ 9a_{22} + 6a_{23} &= 1 \rightarrow a_{22} = 5/9 \\ a_{23} &= -2/3 \end{aligned}$$

3ª coluna:

$$\begin{pmatrix} 4 & 2 & -4 \\ 0 & 9 & 6 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{31} \\ a_{32} \\ a_{33} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{aligned} 4a_{31} + 2a_{32} - 4a_{33} &= 0 \rightarrow a_{31} = 4/3 \\ 9a_{32} + 6a_{33} &= 0 \rightarrow a_{32} = -2/3 \\ a_{33} &= 1 \end{aligned}$$

$$A^{-1} = \begin{pmatrix} 37/18 & -17/18 & 4/3 \\ -17/18 & 5/9 & -2/3 \\ 4/3 & -2/3 & 1 \end{pmatrix} //$$

b)

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & -3 & -1 \\ 2 & 1 & 1 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 2 & 4 & 6 & 1 & 0 & 0 \\ 1 & -3 & -1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 2 & 4 & 6 & 1 & 0 & 0 \\ 0 & -5 & -4 & -1/2 & 1 & 0 \\ 0 & -3 & -5 & -1 & 0 & 1 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|ccc} 2 & 4 & 6 & 1 & 0 & 0 \\ 0 & -5 & -4 & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & -\frac{13}{5} & -\frac{7}{10} & -\frac{3}{5} & 1 \end{array} \right)$$

1ª columna:

$$\begin{pmatrix} 2 & 4 & 6 \\ 0 & -5 & -4 \\ 0 & 0 & -\frac{13}{4} \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{21} \\ a_{31} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{1}{2} \\ -\frac{7}{10} \end{pmatrix} \rightarrow \begin{aligned} 2a_{11} + 4a_{21} + 6a_{31} &= 1 \rightarrow a_{11} = -\frac{1}{13} \\ -5a_{21} - 4a_{31} &= -\frac{1}{2} \rightarrow a_{21} = -\frac{3}{26} \\ -\frac{13}{4}a_{31} &= -\frac{7}{10} \rightarrow a_{31} = \frac{7}{26} \end{aligned}$$

2ª columna:

$$\begin{pmatrix} 2 & 4 & 6 \\ 0 & -5 & -4 \\ 0 & 0 & -\frac{13}{4} \end{pmatrix} \begin{pmatrix} a_{12} \\ a_{22} \\ a_{32} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -\frac{3}{5} \end{pmatrix} \rightarrow \begin{aligned} 2a_{12} + 4a_{22} + 6a_{32} &= 0 \rightarrow a_{12} = \frac{1}{13} \\ -5a_{22} - 4a_{32} &= 1 \rightarrow a_{22} = -\frac{5}{13} \\ -\frac{13}{4}a_{32} &= -\frac{3}{5} \rightarrow a_{32} = \frac{3}{13} \end{aligned}$$

3ª columna:

$$\begin{pmatrix} 2 & 4 & 6 \\ 0 & -5 & -4 \\ 0 & 0 & -\frac{13}{4} \end{pmatrix} \begin{pmatrix} a_{13} \\ a_{23} \\ a_{33} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{aligned} 2a_{13} + 4a_{23} + 6a_{33} &= 0 \rightarrow a_{13} = \frac{7}{13} \\ -5a_{23} - 4a_{33} &= 0 \rightarrow a_{23} = \frac{4}{13} \\ -\frac{13}{4}a_{33} &= 1 \rightarrow a_{33} = -\frac{5}{13} \end{aligned}$$

$$A^{-1} = \begin{pmatrix} -\frac{1}{13} & \frac{1}{13} & \frac{7}{13} \\ -\frac{3}{26} & -\frac{5}{13} & \frac{4}{13} \\ \frac{7}{26} & \frac{3}{13} & -\frac{5}{13} \end{pmatrix}$$

10- a)

$$a = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & -2 & 3 & 4 \\ 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

$$\det(A_1) = 4 \neq 0$$

$$\det(A_2) = 4 \cdot (-1)^2(-2) + 3(-1)^3 \cdot 1 = -1 \neq 0$$

$$\det(A_3) = 4(-1)^2 \begin{vmatrix} -2 & 3 \\ -1 & -1 \end{vmatrix} + 3(-1)^3 \begin{vmatrix} 1 & 3 \\ 1 & -1 \end{vmatrix} + 2(-1)^4 \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix}$$

$$\det(A_3) = 4(-2 \cdot (-1)^2(-1) + 3(-1)^3(-1)) - 3(1(-1)^2(-1) + 3(-1)^3 \cdot 1) + 2(1(-1)^2 \cdot 1 + (-2)(-1)^3 \cdot 1)$$

$$\det(A_3) = 4 \cdot 5 - 3(-4) + 2 \cdot 1 = 34 \neq 0$$

$$\det(A_4) = 4(-1)^2 \begin{vmatrix} -2 & 3 & 4 \\ -1 & -1 & -1 \\ 1 & 1 & 1 \end{vmatrix} + 1(-1)^3 \begin{vmatrix} 3 & 2 & 1 \\ -1 & -1 & -1 \\ 1 & 1 & 1 \end{vmatrix} + 1(-1)^4 \begin{vmatrix} 3 & 2 & 1 \\ -2 & 3 & 4 \\ 1 & 1 & 1 \end{vmatrix}$$

$$+ 1(-1)^5 \begin{vmatrix} 3 & 2 & 1 \\ -2 & 3 & 4 \\ -1 & -1 & -1 \end{vmatrix} = 8 \neq 0$$

$\therefore$ converge

$$\mu_{11} = 4$$

$$L_{21} = 1/4$$

$$\mu_{22} = -11/4$$

$$\mu_{23} = 5/2$$

$$\mu_{24} = 15/4$$

$$\mu_{12} = 3$$

$$L_{31} = 1/4$$

$$L_{32} = 7/11$$

$$\mu_{33} = -34/11$$

$$\mu_{34} = -40/11$$

$$\mu_{13} = 2$$

$$L_{41} = 1/4$$

$$L_{42} = -1/11$$

$$L_{43} = -4/17$$

$$\mu_{44} = 4/17$$

$$\mu_{14} = 1$$

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 1 & 0 & 0 \\ 1/4 & 7/11 & 1 & 0 \\ 1/4 & -1/11 & -4/17 & 1 \end{pmatrix}$$

,

$$U = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 0 & -11/4 & 5/2 & 15/4 \\ 0 & 0 & -34/11 & -40/11 \\ 0 & 0 & 0 & 4/17 \end{pmatrix}$$

$$Ly = b$$

1^{st} column:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 1 & 0 & 0 \\ 1/4 & 7/11 & 1 & 0 \\ 1/4 & -1/11 & -4/17 & 1 \end{pmatrix} \begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{14} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow y = \begin{pmatrix} 1 \\ -1/4 \\ -1/11 \\ -5/17 \end{pmatrix}$$

2^a columna:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 1 & 0 & 0 \\ 1/4 & 7/11 & 1 & 0 \\ 1/4 & -1/11 & -4/17 & 1 \end{pmatrix} \begin{pmatrix} y_{21} \\ y_{22} \\ y_{23} \\ y_{24} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \rightarrow y = \begin{pmatrix} 0 \\ 1 \\ -7/11 \\ -1/17 \end{pmatrix}$$

3^a columna:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 1 & 0 & 0 \\ 1/4 & 7/11 & 1 & 0 \\ 1/4 & -1/11 & -4/17 & 1 \end{pmatrix} \begin{pmatrix} y_{31} \\ y_{32} \\ y_{33} \\ y_{34} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \rightarrow y = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 4/17 \end{pmatrix}$$

4^a columna:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 1 & 0 & 0 \\ 1/4 & 7/11 & 1 & 0 \\ 1/4 & -1/11 & -4/17 & 1 \end{pmatrix} \begin{pmatrix} y_{41} \\ y_{42} \\ y_{43} \\ y_{44} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow y = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$Ux = y$$

1^a columna:

$$\begin{pmatrix} 4 & 3 & 2 & 1 \\ 0 & -1/4 & 5/2 & 15/4 \\ 0 & 0 & -34/11 & -49/11 \\ 0 & 0 & 0 & 4/17 \end{pmatrix} \begin{pmatrix} x_{11} \\ x_{12} \\ x_{13} \\ x_{14} \end{pmatrix} = \begin{pmatrix} 1 \\ -1/4 \\ -1/11 \\ -5/17 \end{pmatrix} \rightarrow x = \begin{pmatrix} 0 \\ -1/4 \\ 3/2 \\ -5/4 \end{pmatrix}$$

2^a columna:

$$\begin{pmatrix} 4 & 3 & 2 & 1 \\ 0 & -1/4 & 5/2 & 15/4 \\ 0 & 0 & -34/11 & -49/11 \\ 0 & 0 & 0 & 4/17 \end{pmatrix} \begin{pmatrix} x_{21} \\ x_{22} \\ x_{23} \\ x_{24} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -7/11 \\ -1/17 \end{pmatrix} \rightarrow x = \begin{pmatrix} 0 \\ -1/4 \\ 1/2 \\ -1/4 \end{pmatrix}$$

3ª columna:

$$\begin{pmatrix} 4 & 3 & 2 & 1 \\ 0 & -1/4 & 5/2 & 15/4 \\ 0 & 0 & -34/11 & -49/11 \\ 0 & 0 & 0 & 4/17 \end{pmatrix} \begin{pmatrix} x_{31} \\ x_{32} \\ x_{33} \\ x_{34} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 4/17 \end{pmatrix} \rightarrow x = \begin{pmatrix} 1/2 \\ 0 \\ -3/2 \\ 1 \end{pmatrix}$$

4ª columna:

$$\begin{pmatrix} 4 & 3 & 2 & 1 \\ 0 & -1/4 & 5/2 & 15/4 \\ 0 & 0 & -34/11 & -49/11 \\ 0 & 0 & 0 & 4/17 \end{pmatrix} \begin{pmatrix} x_{41} \\ x_{42} \\ x_{43} \\ x_{44} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow x = \begin{pmatrix} 1/2 \\ 5/4 \\ -5 \\ 17/5 \end{pmatrix}$$

$$x = \begin{pmatrix} 0 & 0 & 1/2 & 1/2 \\ -1/4 & -1/4 & 0 & 5/4 \\ 3/2 & 1/2 & -3/2 & -5 \\ -5/4 & -1/4 & 1 & 17/2 \end{pmatrix} //$$

b)

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 10 & 2 \\ -1 & 2 & 4 \end{pmatrix}$$

$$\mu_{11} = 2 \quad L_{21} = 1/2 \quad \mu_{22} = 19/2 \quad \mu_{23} = 5/2$$

$$\mu_{12} = 1 \quad L_{31} = -1/2 \quad L_{32} = 5/19 \quad \mu_{33} = 54/19$$

$$\mu_{13} = -1$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -1/2 & 5/19 & 1 \end{pmatrix} ; \quad U = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 19/2 & 5/2 \\ 0 & 0 & 54/19 \end{pmatrix}$$

$$Ly = b$$

1ª columna:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -1/2 & 5/19 & 1 \end{pmatrix} \begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{aligned} y_{11} &= 1 \\ 1/2 y_{11} + y_{12} &= 0 \rightarrow y_{12} = -1/2 \\ -1/2 y_{11} + 5/19 y_{12} + y_{13} &= 0 \rightarrow y_{13} = 12/19 \end{aligned}$$

2ª columna:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -1/2 & 5/19 & 1 \end{pmatrix} \begin{pmatrix} y_{21} \\ y_{22} \\ y_{23} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \rightarrow \begin{aligned} y_{21} &= 0 \\ 1/2 y_{21} + y_{22} &= 1 \rightarrow y_{22} = 1 \\ -1/2 y_{21} + 5/19 y_{22} + y_{23} &= 0 \rightarrow y_{23} = -5/19 \end{aligned}$$

3ª columna:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -1/2 & 5/19 & 1 \end{pmatrix} \begin{pmatrix} y_{31} \\ y_{32} \\ y_{33} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{aligned} y_{31} &= 0 \\ 1/2 y_{31} + y_{32} &= 0 \rightarrow y_{32} = 0 \\ -1/2 y_{31} + 5/19 y_{32} + y_{33} &= 1 \rightarrow y_{33} = 1 \end{aligned}$$

$$Ux = y$$

1ª columna:

$$\begin{pmatrix} 2 & 1 & -1 \\ 0 & 19/2 & 5/2 \\ 0 & 0 & 54/19 \end{pmatrix} \begin{pmatrix} x_{11} \\ x_{12} \\ x_{13} \end{pmatrix} = \begin{pmatrix} 1 \\ -1/2 \\ 12/19 \end{pmatrix} \rightarrow \begin{aligned} 2x_{11} + x_{12} - x_{13} &= 1 \rightarrow x_{11} = 2/3 \\ 19/2 x_{12} + 5/2 x_{13} &= -1/2 \rightarrow x_{12} = -1/9 \\ 54/19 x_{13} &= 12/19 \rightarrow x_{13} = 2/9 \end{aligned}$$

2ª columna:

$$\begin{pmatrix} 2 & 1 & -1 \\ 0 & 19/2 & 5/2 \\ 0 & 0 & 54/19 \end{pmatrix} \begin{pmatrix} x_{21} \\ x_{22} \\ x_{23} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -5/9 \end{pmatrix} \rightarrow \begin{aligned} 2x_{21} + x_{22} - x_{23} &= 0 \rightarrow x_{21} = -1/9 \\ 19/2 x_{22} + 5/2 x_{23} &= 1 \rightarrow x_{22} = 7/54 \\ 54/19 x_{23} &= -5/9 \rightarrow x_{23} = -5/54 \end{aligned}$$

3ª coluna:

$$\begin{pmatrix} 2 & 1 & -1 \\ 0 & 19/2 & 5/2 \\ 0 & 0 & 54/19 \end{pmatrix} \begin{pmatrix} x_{31} \\ x_{32} \\ x_{33} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{aligned} 2x_{31} + x_{32} - x_{33} &= 0 \rightarrow x_{31} = 2/9 \\ 19/2 x_{32} + 5/2 x_{33} &= 0 \rightarrow x_{32} = -5/54 \\ 54/19 x_{33} &= 1 \rightarrow x_{33} = 19/54 \end{aligned}$$

$$x = \begin{pmatrix} 2/9 & -1/9 & 2/9 \\ -1/9 & 7/54 & -5/54 \\ 2/9 & -5/54 & 19/54 \end{pmatrix} //$$

11-a) $5x_1 + 2x_2 + x_3 = 7$

$$-x_1 + 4x_2 + 2x_3 = 3$$

$$2x_1 - 3x_2 + 10x_3 = -1$$

i) $|5| > |2| + |1|$

$$|4| > |-1| + |2|$$

$$|10| > |2| + |-3|$$

ii) critério das colunas:

$$j=1 \rightarrow \left| \frac{2}{5} \right| + \left| \frac{1}{5} \right| = \frac{3}{5}$$

$$j=2 \rightarrow \left| \frac{-1}{4} \right| + \left| \frac{2}{4} \right| = \frac{3}{4}$$

$$j=3 \rightarrow \left| \frac{2}{10} \right| + \left| \frac{-3}{10} \right| = \frac{1}{2}$$

$$\max \left\{ \left| \frac{3}{5}, \frac{3}{4}, \frac{1}{2} \right| \right\}$$

$$= \frac{3}{4} < 1 \quad \therefore \text{converge}$$

iii)

$$x_1 = \frac{7 - 2x_2 - x_3}{5}, \quad x_2 = \frac{3 + x_1 - 2x_3}{4}, \quad x_3 = \frac{-1 - 2x_1 + 3x_2}{10}$$

$$K=0 \quad x_1^{(1)} = \frac{7 - 2(3,9) - 0,2}{5} = -0,2$$

$$x_2^{(1)} = \frac{3 + (-1,4) - 2(0,2)}{4} = 0,3$$

$$x_3^{(1)} = \frac{-1 - 2(-1,4) + 3(3,9)}{10} = 1,35$$

$$\|x^{(1)}\|_{\infty} = \max\{|-0,2|, |0,3|, |1,35|\} = 1,35$$

$$\begin{aligned} \|x^{(1)} - x^{(0)}\| &= \max\{|-0,2 + 1,4|, |0,3 - 3,9|, |1,35 - 0,2|\} \\ &= \max\{|1,2|, |-3,6|, |1,15|\} = 3,6 \end{aligned}$$

$$cp = \frac{3,6}{1,35} = 2,67 > \varepsilon$$

$$K=1 \quad x_1^{(2)} = \frac{7 - 2(0,3) - 1,35}{5} = 1,01$$

$$x_2^{(2)} = \frac{3 + (-0,2) - 2(1,35)}{4} = 0,025$$

$$x_3^{(2)} = \frac{-1 - 2(-0,2) + 3(0,3)}{10} = 0,03$$

$$\|x^{(2)}\|_{\infty} = \max\{|1,01|, |0,025|, |0,03|\} = 1,01$$

$$\begin{aligned} \|x^{(2)} - x^{(1)}\|_{\infty} &= \max\{|1,01 + 0,2|, |0,025 - 0,3|, |0,03 - 1,35|\} \\ &= \max\{|1,21|, |-0,275|, |-1,32|\} = 1,32 \end{aligned}$$

$$cp = \frac{1,32}{1,01} = 1,31 > \varepsilon$$

$$K=2 \quad x_1^{(3)} = \frac{7 - 2(0,025) - 0,03}{5} = 1,384$$

$$x_2^{(3)} = \frac{3 + 1,01 - 2(0,23)}{4} = 0,9875$$

$$x_3^{(3)} = \frac{-1 - 2(1,01) + 3(0,025)}{10} = -0,2945$$

$$\|x^{(3)}\|_{\infty} = \max\{|1,384|, |0,9875|, |-0,2945|\} = 1,384$$

$$\begin{aligned} \|x^{(3)} - x^{(2)}\|_{\infty} &= \max\{|1,384 - 1,01|; |0,9875 - 0,025|; |-0,2945 - 0,03|\} \\ &= \max\{|0,374|; |0,9625|; |-0,3245|\} = 0,9625 \end{aligned}$$

$$cp = \frac{0,9625}{1,384} = 0,6954 > \varepsilon$$

$$K=3 \quad x_1^{(4)} = \frac{7 - 2(0,9875) + 0,2945}{5} = 1,0639$$

$$x_2^{(4)} = \frac{3 + 1,384 - 2(-0,2945)}{4} = 1,24325$$

$$x_3^{(4)} = \frac{-1 - 2(1,384) + 3(0,9875)}{10} = -0,08055$$

$$\|x^{(4)}\|_{\infty} = \max\{|1,0639|; |1,24325|; |-0,08055|\} = 1,24325$$

$$\begin{aligned} \|x^{(4)} - x^{(3)}\|_{\infty} &= \max\{|1,0639 - 1,384|; |1,24325 - 0,9875|; |-0,08055 - 0,2945|\} \\ &= \max\{|-0,3201|; |0,25575|; |-0,21395|\} = 0,3201 \end{aligned}$$

$$cp = \frac{0,3201}{1,24325} = 0,2575 > \varepsilon$$

$$K=4 \quad x_1^{(5)} = \frac{7 - 2(1,24325) - 0,08055}{5} = 0,9188$$

$$x_2^{(5)} = \frac{3 + 1,0639 - 2(-0,08055)}{4} = 1,0563$$

$$x_3^{(5)} = \frac{-1 - 2(1,0639) + 3(1,24325)}{10} = 0,0602$$

$$\|x^{(5)}\|_{\infty} = \max\{|0,9188|; |1,0563|; |0,0602|\} = 1,0563$$

$$\begin{aligned} \|x^{(5)} - x^{(4)}\|_{\infty} &= \max\{|0,9188 - 1,0639|; |1,0563 - 1,24325|; |0,0602 + 0,08055|\} \\ &\quad \max\{|-0,1451|; |-0,18695|; |0,14075|\} = 0,18695 \end{aligned}$$

$$cp = \frac{0,18695}{1,0563} = 0,1769 > \varepsilon$$

$$K=5 \quad x_1^{(6)} = \frac{7 - 2(1,0563) - 0,0602}{5} = 0,9655$$

$$x_2^{(6)} = \frac{3 + 0,9188 - 2(0,0602)}{4} = 0,9496$$

$$x_3^{(6)} = \frac{-1 - 2(0,9188) + 3(1,0563)}{10} = 0,0331$$

$$\|x^{(6)}\|_{\infty} = \max\{|0,9655|; |0,9496|; |0,0331|\} = 0,9655$$

$$\begin{aligned} \|x^{(6)} - x^{(5)}\| &= \max\{|0,9655 - 0,9188|; |0,9496 - 1,0563|; |0,0331 - 0,0602|\} \\ &\quad \max\{|0,0467|; |-0,1067|; |-0,0271|\} = 0,1067 \end{aligned}$$

$$cp = \frac{0,1067}{0,9655} = 0,1105 > \varepsilon$$

$$K=6 \quad x_1^{(7)} = \frac{7 - 2(0,9496) - 0,0331}{5} = 1,0135$$

$$x_2^{(7)} = \frac{3 + (0,9655) - 2(0,0331)}{4} = 0,9748$$

$$x_3^{(7)} = \frac{-1 - 2(0,9655) + 3(0,9496)}{10} = -0,0082$$

$$\|x^{(7)}\|_{\infty} = \max\{|1,0135|; |0,9748|; |-0,0082|\} = 1,0135$$

$$\|x^{(7)} - x^{(6)}\|_{\infty} = \max\{|1,0135 - 0,9655|; |0,9748 - 0,9496|; |-0,0082 - 0,0331|\} \\ \max\{|0,0481|; |0,0252|; |-0,0413|\} = 0,048$$

$$cp = \frac{0,048}{1,0135} = 0,0474 < \varepsilon$$

$$x \approx \begin{bmatrix} 1,0135 \\ 0,9748 \\ -0,0081 \end{bmatrix} //$$

$$b) \quad 10x_1 + x_2 + x_3 = 10$$

$$2x_1 + 10x_2 + 8x_3 = 20$$

$$7x_1 + x_2 + 10x_3 = 20$$

$$i) \quad |10| > |1| + |1|$$

$$|10| > |2| + |8| \rightarrow |10| = |10|$$

$\therefore$ ainda pode convergir

$$|10| > |7| + |1|$$

ii) critério das colunas:

$$j=1 \rightarrow \left| \frac{2}{10} \right| + \left| \frac{7}{10} \right| = \frac{9}{10}$$

$$j=2 \rightarrow \left| \frac{1}{10} \right| + \left| \frac{1}{10} \right| = \frac{1}{5}$$

$$\max \left\{ \frac{9}{10}, \frac{1}{5}, \frac{9}{10} \right\} = \frac{9}{10} < 1$$

$$j=3 \rightarrow \left| \frac{1}{10} \right| + \left| \frac{8}{10} \right| = \frac{9}{10}$$

$\therefore$ converge

$$\text{iii)} \quad x_1 = \frac{10 - x_2 - x_3}{10}, \quad x_2 = \frac{20 - 2x_1 - 8x_3}{10}, \quad x_3 = \frac{20 - 7x_1 - x_2}{10}$$

$$K=0 \quad x_1^{(1)} = \frac{10 - 1 - 1}{10} = 0,8$$

$$x_2^{(1)} = \frac{20 - 2 - 8}{10} = 1$$

$$x_3^{(1)} = \frac{20 - 7 - 1}{10} = 1,2$$

$$K=1 \quad x_1^{(2)} = \frac{10 - 1 - 1,2}{10} = 0,78; \quad x_2^{(2)} = \frac{20 - 2(0,8) - 8(1,2)}{10} = 0,88$$

$$x_3^{(2)} = \frac{20 - 7(0,8) - 1}{10} = 1,34$$

$$K=2 \quad x_1^{(3)} = \frac{10 - 0,88 - 1,34}{10} = 0,778; \quad x_2^{(3)} = \frac{20 - 2(0,78) - 8(1,34)}{10} = 0,772$$

$$x_3^{(3)} = \frac{20 - 7(0,78) - 0,88}{10} = 1,366$$

$$K=3 \quad x_1^{(4)} = \frac{10 - 0,772 - 1,366}{10} = 0,7862; \quad x_2^{(4)} = \frac{20 - 2(0,778) - 8(1,366)}{10} = 0,7516$$

$$x_3^{(4)} = \frac{20 - 7(0,778) - 0,772}{10} = 1,3782$$

$$K=4 \quad x_1^{(5)} = \frac{10 - 0,7516 - 1,3782}{10} = 0,78702$$

$$x_2^{(5)} = \frac{20 - 2(0,7862) - 8(1,3782)}{10} = 0,7402; \quad x_3^{(5)} = \frac{20 \cdot 7(0,7862) - 0,7516}{10} = 1,3745$$

$$\|x^{(5)}\|_{\infty} = \max\{|0,78702|; |0,7402|; |1,3745|\} = 1,3745$$

$$\|x^{(5)} - x^{(4)}\| = \max\{|0,78702 - 0,7862|; |0,7402 - 0,7516|; |1,3745 - 1,3782|\} \\ \max\{|0,00082|; |-0,0114|; |-0,0037|\} = 0,0114$$

$$cp = \frac{0,0114}{1,3745} = 0,0083 < \varepsilon$$

$$x = \begin{bmatrix} 0,78702 \\ 0,7402 \\ 1,3745 \end{bmatrix}$$

$$12-a) \quad 4x_1 + x_2 + x_3 + x_4 = 7$$

$$2x_1 - 8x_2 + x_3 - x_4 = -6$$

$$x_1 + 2x_2 - 5x_3 + x_4 = -1$$

$$x_1 + x_2 + x_3 - 4x_4 = -1$$

$$x^{(0)} = \begin{pmatrix} 0 \\ 0,2 \\ -0,4 \\ 0 \end{pmatrix} \quad ; \quad \varepsilon = 10^{-2}$$

$$i) \quad a_{ii} \neq 0$$

$$4 \neq 0, -8 \neq 0, -5 \neq 0, -4 \neq 0$$

$$ii) \quad \det \begin{vmatrix} 4 & 1 & 1 & 1 \\ 2 & -8 & 1 & -1 \\ 1 & 2 & -5 & 1 \\ 1 & 1 & 1 & -4 \end{vmatrix} = -755 \neq 0$$

iii) critério de Sassenfeld

$$\beta_1 = \frac{1+1+1}{4} = \frac{3}{4} ; \quad \beta_2 = \frac{2 \cdot \frac{3}{4} + 1 + 1}{8} = \frac{3,5}{8} = 0,4375$$

$$\beta_3 = \frac{\frac{3}{4} + 2 \cdot \frac{3,5}{8} + 1}{5} = \frac{2,625}{5} = 0,525 ; \quad \beta_4 = \frac{\frac{3}{4} + \frac{3,5}{8} + \frac{2,625}{5}}{4} = \frac{1,7125}{4} = 0,4281$$

$$\beta_{\max} = \frac{3}{4} < 1 \quad \therefore \text{converge}$$

$$\text{iv)} \quad x_1 = \frac{7 - x_2 - x_3 - x_4}{4} ; \quad x_2 = \frac{6 + 2x_1 + x_3 - x_4}{8}$$

$$x_3 = \frac{1 + x_1 + 2x_2 + x_4}{5} ; \quad x_4 = \frac{1 + x_1 + x_2 + x_3}{4}$$

$$K=0 \quad x_1^{(1)} = \frac{7 - 0,2 + 0,4 - 0}{4} = 1,8 ; \quad x_2^{(1)} = \frac{6 + 2(1,8) - 0,4 - 0}{8} = 1,15$$

$$x_3^{(1)} = \frac{1 + 1,8 + 2(1,15) + 0}{5} = 1,02 ; \quad x_4^{(1)} = \frac{1 + 1,8 + 1,15 + 1,02}{4} = 1,2425$$

$$K=1 \quad x_1^{(2)} = \frac{7 - 1,15 - 1,02 - 1,2425}{4} = 0,8969$$

$$x_2^{(2)} = \frac{6 + 2(0,896875) + 1,02 - 1,2425}{8} = 0,9464$$

$$x_3^{(2)} = \frac{1 + 0,896875 + 2(0,94640625) + 1,2425}{5} = 1,0064$$

$$x_4^{(2)} = \frac{1 + 0,896875 + 0,94640625 + 1,0064375}{4} = 0,9624$$

$$K=2 \quad x_1^{(3)} = \frac{7 - 0,9464 - 1,0064 - 0,9624}{4} = 1,0212$$

$$x_2^{(3)} = \frac{6 + 2(1,0212) + 1,0064 - 0,9624}{8} = 1,0585$$

$$x_3^{(3)} = \frac{1 + 1,0212 + 2(1,0585) + 0,9624}{5} = 1,0201$$

$$x_4^{(3)} = \frac{1 + 1,0212 + 1,0585 + 1,0201}{4} = 1,0249$$

$$K=3 \quad x_1^{(4)} = \frac{7 - 1,0585 - 1,02012 - 1,0249}{4} = 0,9741$$

$$x_2^{(4)} = \frac{6 + 2(0,9741) + 1,0201 - 1,0249}{8} = 0,9929$$

$$x_3^{(4)} = \frac{1 + 0,9741 + 2(0,9929) + 1,0249}{5} = 0,9969$$

$$x_4^{(4)} = \frac{1 + 0,9741 + 0,9929 + 0,9970}{4} = 0,9910$$

$$K=4 \quad x_1^{(5)} = \frac{7 - 0,9929 - 0,9970 - 0,991}{4} = 1,0048$$

$$x_2^{(5)} = \frac{6 + 2(1,0048) + 0,9970 - 0,991}{5} = 1,0020$$

$$x_3^{(5)} = \frac{1 + 1,0048 + 2(1,0080) + 0,991}{5} = 1$$

$$x_4^{(5)} = \frac{1 + 1,0048 + 1,0020 + 1,000}{4} = 1,0017$$

$$K=5 \quad x_1^{(6)} = \frac{7 - 1,0020 - 1 - 1,0017}{4} = 0,9991$$

$$x_2^{(6)} = \frac{6 + 2(0,9991) + 1 - 1,0017}{4} = 0,9999$$

$$x_3^{(6)} = \frac{1 + 0,9991 + 2(0,9999) + 1,0017}{5} = 1,00012$$

$$x_4^{(6)} = \frac{1 + 0,9991 + 0,9999 + 1,0001}{4} = 0,9998$$

$$\|x^{(6)}\|_{\infty} = 1,0001$$

$$\|x^{(6)} - x^{(5)}\| = \max\{|-0,0057|; |-0,0001|; |0,0001|; |-0,0019|\} = 0,0057$$

$$cp = \frac{0,0057}{1,0001} = 0,006 < \varepsilon$$

$$x = \begin{bmatrix} 0,9991 \\ 0,9999 \\ 1,0001 \\ 0,9998 \end{bmatrix} //$$

$$b) \quad 4x_1 + x_2 + x_3 = 6$$

$$x_1 + 6x_2 + x_3 = 8$$

$$2x_1 + x_2 + 8x_3 = 11$$

$$x^{(0)} = \begin{pmatrix} 0 \\ 0,1 \\ 0 \end{pmatrix}, \quad \varepsilon = 10^{-2}$$

$$i) \quad a_{ii} \neq 0$$

$$4 \neq 0, 6 \neq 0, 8 \neq 0$$

$$ii) \quad \det A \neq 0$$

$$\det \begin{vmatrix} 4 & 1 & 1 \\ 1 & 6 & 1 \\ 2 & 1 & 8 \end{vmatrix} = 171 \neq 0$$

iii) Critério de Sassenfeld

$$\beta_1 = \frac{1+1}{4} = \frac{1}{2} ; \quad \beta_2 = \frac{1+1}{6} = \frac{1}{3} ; \quad \beta_3 = \frac{2+1}{8} = \frac{3}{8}$$

$$\beta_{\max} = \frac{1}{2} < 1 \quad \therefore \text{converge}$$

iv)

$$x_1 = \frac{6 - x_2 - x_3}{4} ; \quad x_2 = \frac{8 - x_1 - x_3}{6} ; \quad x_3 = \frac{11 - 2x_1 - x_2}{8}$$

$$K=0 \quad x_1^{(1)} = \frac{6 - 0 - 0}{4} = 1,475$$

$$x_2^{(1)} = \frac{8 - 1,475 - 0}{6} = 1,0875$$

$$x_3^{(1)} = \frac{11 - 2(1,475) - 1,0875}{8} = 0,8703$$

$$K=1 \quad x_1^{(2)} = \frac{6 - 1,0875 - 0,8703}{4} = 1,0106$$

$$x_2^{(2)} = \frac{8 - 1,0106 - 0,8703}{6} = 1,0199$$

$$x_3^{(2)} = \frac{11 - 2(1,0106) - 1,0199}{8} = 0,9949$$

$$K=2 \quad x_1^{(3)} = \frac{6 - 1,0199 - 0,9949}{4} = 0,9963$$

$$x_2^{(3)} = \frac{8 - 0,9963 - 0,9949}{6} = 1,0015$$

$$x_3^{(3)} = \frac{11 - 2(0,9963) - 1,0015}{8} = 1,0007$$

$$K=3 \quad x_1^{(4)} = \frac{6 - 1,0015 - 1,0007}{4} = 0,9995$$

$$x_2^{(4)} = \frac{8 - 0,9995 - 1,0007}{6} = 1$$

$$x_3^{(4)} = \frac{11 - 2(0,9995) - 1}{8} = 1,0001$$

$$\|x^{(4)}\|_{\infty} = \max\{|0,9995|, |1|, |1,0001|\} = 1,0001$$

$$\|x^{(4)} - x^{(3)}\| = \max\{|0,0032|, |1 - 0,0015|, |1 - 0,0006|\} = 0,0032$$

$$\rho = \frac{0,0032}{1,0001} = 0,0032 < \varepsilon$$

$$x = \begin{bmatrix} 0,9995 \\ 1 \\ 1,0001 \end{bmatrix} //$$